Fiscal Year:	FY 2019	**Task Last Updated:**	FY 09/13/2018
PI Name:	Oates, Chris Ph.D.		
Project Title:	NIST/ACES Collaboration: ACES Ground Station and Data Analysis Center in Boulder, Colorado		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Fundamental physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
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Space Biology Special Category:	None		
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Zip Code:	80305-3337	**Congressional District:**	2
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2011 Fundamental Physics NNH11ZTT001N
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No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
Contact Monitor:	Israelsson, Ulf	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 3/31/2021 due to change in Principal Investigator, per PI and U. Israelsson/JPL (Ed., 9/13/21) NOTE: End date changed to 12/31/2022 per U. Israelsson/JPL (Ed., 9/13/18) NOTE: End date changed to 11/30/2018 per M. Lee/NASA HQ (Ed., 2/16/16)		
Key Personnel Changes/Previous PI:	EDITOR'S NOTE (December 2015): PI Chris Oates took over the project from Thomas O'Brian on 11/16/2015.		
COI Name (Institution):	Jefferts, Steven Ph.D. (National Institute of Standards & Technology)		
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Task Description:

EDITOR'S NOTE (December 2015): Principal Investigator (PI) Chris Oates took over the project from Thomas O'Brian on 11/16/2015.
To support primary Atomic Clock Ensemble in Space (ACES) mission objectives, the National Institute of Standards and Technology (NIST) proposes to:

1. Develop a Microwave Link Ground Terminal (MWL GT) at NIST in Boulder, CO to perform 10-17 uncertainty time and frequency comparisons between ACES clocks and world-leading optical and atomic clocks at NIST Boulder and the nearby JILA (formerly known as the Joint Institute for Laboratory Astrophysics); and to support general ACES science goals.

2. Participate as fully as possible in coordinated European Space Agency (ESA) data analysis with formal ESA Data Analysis Centers and other ACES partners throughout the world. NIST will collaborate closely with ESA partners to define how NIST can best contribute to overall ACES mission goals within the constraints of NIST resources.

As part of participation in the time and frequency transfer and data analysis, NIST will participate in tests of fundamental physics based on ACES data to include at least the following:

• Global time and frequency transfer with uncertainty as low as 10-17.

• Direct comparison of PHARAO space clock to NIST time and frequency standards at uncertainty of 10-17 or better.

• Test of gravitation redshift at 2 x 10-6 uncertainty or better.

• Search for variation in fundamental constants, such as fine structure constant at level of 10-18 per year or better.

All these goals will be facilitated by the strong time and frequency infrastructure at NIST, including (a partial list):

• World-class cesium microwave primary frequency standards, including NIST-F1 with fractional frequency uncertainty of 3 x 10-16, and NIST-F2 which will be operational in 2013 with an eventual uncertainty of 1 x 10-16 or better.

• Two mercury ion optical clocks with current fractional frequency uncertainty of approximately 1 x 10-17 and continually improving.

• Two aluminum ion “logic” optical clocks with current fractional frequency uncertainty of approximately 8 x 10-18 and continually improving.

• Two ytterbium lattice optical clocks with uncertainties better than 1 x 10-16 and rapidly improving.

• Two strontium lattice optical clocks with uncertainties approximately 3 x 10-17 and rapidly improving. (At JILA. A dedicated fiber optic system connects NIST and JILA to enable the JILA clocks to directly participate in NIST time and frequency transfer and intercomparisons.)

• A high performance calcium optical frequency standard with exquisite short-term stability to serve as a “fly wheel” for clock comparisons.

• A world-class time scale comprising seven hydrogen masers, with eight more masers to be added over the next three years.

NIST scientists will participate in ESA/NASA planning meetings about the ACES program in general, and will specifically consult with ESA to determine how NIST can most fully participate in MLW GT data analysis and applications.

In addition to the NIST resources already proposed for this program (NIST and JILA optical and microwave frequency standards, NIST time scale, NIST time and frequency transfer capabilities, etc.) NIST will directly contribute substantial funds and expert NIST staff time significantly exceeding the direct NASA investment.

We propose the following revised program plan outline:

Year 1 (2012): Begin design and construction of platform and other infrastructure for MWL hardware. $150k for partial payment of platform and related infrastructure design and construction.

Year 2 (2013): Complete construction of platform and other infrastructure for MWL system. $150k for partial payment of platform and related infrastructure.

Year 3 (2014): Receive MWL hardware. Install, test, and integrate MWL hardware into NIST time and frequency transfer system. Prepare and test MWL system for ACES time and frequency transfer. New NIST staff to prepare for participation in MWL data analysis. $150k for partial support of staff time.

Year 4 (2015): ACES launch. Participate in ACES time and frequency transfer, data analysis, and tests of fundamental physics. $150k for partial support of staff time.

Year 5 (2016): On-going ACES experiments. Participate in ACES time and frequency transfer, data analysis, and tests of fundamental physics. $150k for partial support of staff time.

Gap Year (2017): Perform relativistic calculations to prevent unanticipated shifts from contaminating ACES data. Install updated MWL hardware into new NIST time and frequency transfer system. Prepare updated data analysis system in anticipation of ACES launch. $150k for partial support of staff time.

Gap 6 Mo. (2018): Assuming successful launch, participate in ACES time and frequency transfer, data analysis, and tests of fundamental physics. $80k for partial support of staff time.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

The major benefits and impacts of this research program can be grouped into three main areas:
1. Substantial improvements in international coordination of extremely precise and accurate time and frequency information. Precision timekeeping and synchronization are crucial for such ubiquitous technologies as precision positioning and navigation, telecommunications, remote sensing, surveillance, and many other applications. The current technical limitations on international time and frequency transfer impose fundamental limits on these technologies. This project seeks to improve international time and frequency transfer by at least a factor of 10, with the potential for significantly greater improvements. Such improvements would not only directly enhance the current technologies depending on precision timing and synchronization, but would also open up new applications directly related to ultraprecise timing such as gravimetry, geodesy, magnetometry, and other measurements based on tiny perturbations in precision frequency measurements.

2. This project will serve as a pathfinder demonstration of the ability to perform cold atom experiments in space. Cold atoms and molecules represent a major, growing research opportunity for a wide variety of technologies, from ultraprecision measurements of all kinds (time, gravity, magnetic fields, electrical quantities, temperature, etc.) to the potential for development of exotic new materials. Having mastered cold atom and molecule experiments in the Earth laboratory, being able to perform such experiments in space opens the benefits of precision measurements in the space environment as well as microgravity experiments not possible in the Earth-based laboratory.

3. Fundamental physics experiments. The operation of a cold atom clock in space provides the opportunity to significantly improve measurements on the limits of special and general relativity and related phenomena, such search for possible Lorentz transformation variations and search for possible time variations in fundamental constants, at levels 10 to 100 times more precise than previous experiments. Most major physics discoveries in recent decades have come from tiny variations in expected behavior, and these experiments would have profound implications on fundamental physics and cosmology.

Task Progress:

In the most recent reporting period (November 2017 through November 2018) work has focused on three main tasks:
1. Installation and evaluation of a high-precision timing link between the to-be-installed Microwave Link Ground Terminal (MWL GT) and NIST's Time Scale, UTC (NIST)

2. Advancing the NIST time and frequency infrastructure time and frequency metrology infrastructure in support of ACES goals for precision time and frequency transfer and tests of fundamental physics

3. Preparing the data analysis infrastructure and understanding the trajectory of the International Space Station

Progress on each task is described below.

1. Installation and evaluation of a high-precision timing link between the to-be-installed Microwave Link Ground Terminal (MWL GT) and NIST's Time Scale, UTC (NIST)

The basis for the ACES program will be the advanced primary frequency standard on the International Space Station, which will link to participating microwave ground stations around the world. NIST has yet to receive its Microwave Link Ground Terminal (MWL GT) from ESA and is awaiting confirmation of an expected delivery date. NIST has been closely coordinating with ESA on details of the MWL GT hardware, information technology and infrastructure requirements.

In the meantime NIST clock scientists have been putting in place the required infrastructure for connecting the ACES clock signals to NIST's most stable microwave sources. Indeed, the main focus over the past year was to construct such a link and demonstrate performance sufficient to support time transfer beyond that of the respective timing sources. We have transferred UTC (NIST), NIST's Time Scale clock signal, from its reference point in Building 1 to the roof of Building 81 (where a constructed platform awaits the arrival and installation of the MGT) through carefully-calibrated "White Rabbit" links. (White Rabbit is a CERN-develop network time transfer protocol that enables sub-nanosecond accuracy for short distance time transfer.) Included in this design are additional monitoring systems for maintaining and preserving the integrity of the transferred 1 PPS and 10 MHz signals.

We have performed extensive measurements on this system to evaluate its performance over one hour, one day, one month, and one year time periods. We find that by locking the output of an H maser (located within 20 m of the MWL GT) to UTC (NIST) via the White Rabbit link, we can surpass the ACES requirements for averaging times up to 1000 s. On longer time scales, the White Rabbit signal itself will meet ACES requirements. As a final check for robustness, we ran the link for the duration of a year and found no degredation. In short, we have demonstrated a hardware link that is fully capable of supporting the ACES-UTC (NIST) comparisons.

Finally, we installed much of the data transfer system including data cables and computer hardware. We have also set up a dedicated computer for data transfer and archiving between the NIST UTC time scale and the ACES MWL terminal, with software that is flexible in its ability to accept a variety of query forms, including the XML format requested by ACES. Included in this development has been the required IT security protocols. These have been tested and should not impede system performance.

2. Advancing the NIST time and frequency infrastructure time and frequency metrology infrastructure in support of ACES goals for precision time and frequency transfer and tests of fundamental physics

NIST continues to improve the performance of its multiple optical and microwave frequency standards, which will be crucial to international time and frequency comparisons and ACES-mediated fundamental research. The past project year included landmark advances in several of NIST's frequency standards.

The NIST ytterbium lattice optical clocks have published a final uncertainty budget with a fractional frequency uncertainty to below 1.4 x 10^-18, a level they can now reach in a few hours of averaging time. A month-long comparison between the two Yb clocks demonstrated agreement at the 7 x 10^-19 level, more than a factor of five beyond current state-of-the-art reproducibility. Taking advantage of the Yb clock's world leading stability, the comparison also reached a record precision of 3 x 10^-19. This remarkable stability dramatically advances the goals of ACES time and frequency transfer given the relatively short visibility of the International Space Station (ISS) to any MWL GT.

The NIST Al+ optical clock has reduced its major source of uncertainty, time dilation shifts due to micromotion, to a

	less than 1 part in 10^-18, fractionally. As a result, the team was able to reduce the total uncertainty budget for the Al+ clock to less than 1 x 10^-18, the lowest clock uncertainty to date worldwide. NIST completed its first set of measurements between the NIST Time Scale and its optical standards, which led to an absolute measurement of the Yb clock transition frequency in terms of Cs, the basis of the definition of the SI second. This measurement had an uncertainty of < 3 x 10^-16, the lowest for any absolute frequency, and one that is basically limited by the Cs standards themselves. A simulation that uses this same data to steer the NIST Time Scale indicated that an improvement of 3-5 would be possible for the Time Scale using the sporadic Yb running schedule already available today. At the time of this report, NIST (including NIST/JILA) currently operates two of the best microwave primary frequency standards (Cs fountains) and four leading optical standards (Yb and Sr neutral atom lattice clocks, Al+ ion clocks, and a Ca thermal beam clock). Other laboratories across the world are achieving high performance as well, and the NIST optical frequency standards are likely to continue to improve. This dramatic improvement of optical frequency standard performance strongly supports ACES goals of improved international time and frequency transfer and the ability to improve fundamental tests of physics. Recently, we have completed the first set of frequency ratio measurements between the optical frequency standards (Yb, Sr, Al+) with uncertainties well below 1 x 10^-17, which have also enabled tests of drifts of fundamental constants. 3. Preparing the data analysis infrastructure and understanding the trajectory of the International Space Station A critical part of our preparation for the ACES-mediated clock comparisons is a consistent handling of the data sets by all participating laboratories. In May 2018 we received a large set of ACES data simulation files from Jet Propulsion Laboratory, which enabled us to clear up some ambiguities seen in previous files. We now concur with the formalism developed by the ACES team for computing the difference in the proper time between the ISS and MWL terminal (desynchronization). Using these data we plan to constrain the gravitational redshift within limits using the accurately determined position and velocity of the ISS. During the reporting period we examined more closely the acceleration noise for clocks in orbit. To mimic an optimal (noise free) environment, we generated a mock data set comprising actual fractional frequency data from comparisons between NIST H-masers and the Cs-fountain NIST F-1. All the systematic effects such as maser drifts were accounted for by fitting the data with a polynomial—leaving only the gravitational signal that we sought to study. We modified the data to take into account the actual orbit of the ISS, which takes the space station over the NIST, Boulder location for periods of a few hundred seconds. We then used the mock data to study the effect of the white acceleration noise after canceling of common mode noise present in both H-maser and Cs-tube onboard the ISS. It can be shown that the rms displacement in position of an object subject to white acceleration noise in one direction for a time T is on the average proportional to T^3/2, with a coefficient depending on the spectral density of the noise. As a result, and the degradation in clock performance due to acceleration noise can be as high as a factor of 10 for a net 10 m displacement.
Bibliography Type:	Description: (Last Updated: 04/05/2023)
Papers from Meeting Proceedings	Ascarrunz FG, Yudin YO, Delgado-Aramburo MC, Ascarrunz LI, Savory J, Banducci A, Jefferts SR. "A portable 87Rb atomic clock with frequency instability at one day in the 10^-15 range." Presented at the 2018 IEEE International Frequency Control Symposium, Squaw Valley, CA, May 21-24, 2018. 2018 IEEE International Frequency Control Symposium, Squaw Valley, CA, May 21-24, 2018. https://doi.org/10.13140/RG.2.2.29820.44160 , May-2018
Papers from Meeting Proceedings	Ascarrunz FG, Yudin YO, Delgado-Aramburo MC, Savory J, Jefferts SR. "Long-term frequency instability of a portable cold 87Rb atomic clock." Presented at 2018 Precision Time and Time Interval Meeting, Reston, VA, January 29-31, 2018. 2018 Precision Time and Time Interval Meeting, Reston, VA, January 29-31, 2018. To be printed in meeting proceedings. https://doi.org/10.13140/RG.2.2.36607.92328 , Jan-2018